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SUPERSEDING
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FEDERAL SPECIFICATION

TRANSIT, ONE-MINUTE; AND TRANSIT TRIPODS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one-minute transits and tripods for use with transits.

1.2 Classification.

1.2.1 Classes. Transits shall be of the following classes, as specified (see 6.2):

Class A - Small, lightweight; mountain or mining transits.
Class B - Average size and weight; engineer's and city transits.
Class C - Maximum size and weight; surveyor's traverse and triangulation transits.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

GG-M-95	Magnifier and Reducing, Glass
TT-E-522	Enamel, Phenolic, Outside
GGG-P-501	Plumb Bob

FEDERAL STANDARDS

FED-STD-H28	Screw Threads for Federal Services.
FED-STD-123	Marking for Shipment (Civil Agencies).

Beneficia1 comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center/MMEDO, Kelly AFB, Texas 78241 by using tile self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearin at the end of this document or by letter,

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MILITARY SPECIFICATIONS

MIL-C-496	Clip, End Strap
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543	Buckle, Tongueless and Web Strap
MIL-C-675	Coating of Glass Optical Elements (Anti-reflection)
MIL-T-704	Treatment and Painting of Materiel
MIL-R-3390	Ring, DEE
MIL-S-3534	Surveying Instruments and Accessories, Packaging of
MIL-G-10924	Grease Automotive and Artillery
MIL-S-43770	Snap Hooks, General Specification For.

MILITARY STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized Unit Loads
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D-3951-82 Packaging, Commercial

{Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 Riverside Plaza, Chicago, IL 60606.)

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National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, 1616 P. Street, N.W., Washington, D.C. 20036.)

National Bureau of Standards

Circular 428	A Test of Lens Resolution for the Photographer.
Circular 533	Method for Determining the Resolving Power of Photographic Lenses.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The transits shall consist essentially of an internally focusing telescope, horizontal and vertical circles and verniers, compass, two plate levels and telescope level, standards, bearings, clamps, and tangent screws, leveling screws, and base plate. Transits shall conform to the requirements shown in Table 1. Unless otherwise specified (see 6.2), extension leg tripods of either the American or European type shall be furnished.

3.2 Preproduction model. The preproduction model is defined as a production model of the product which the manufacturer will subsequently produce in fulfillment of the contract. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a preproduction model for examination and tests to determine conformance to this specification. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations of production units from the preproduction model shall be subject to the approval of the contracting officer.

3.3 Material. Material shall be as specified herein. The materials used shall be corrosion-resistant and nonmagnetic. Materials shall be such as to insure the same coefficient of expansion within practicable limits between all bearing surfaces. Castings shall be of virgin metals, smooth and free from blowholes, sandholes, and without plugging. Bearings shall be made from sound pieces without holes and without plugging. When a definite material is not specified, a material which will enable the transit to meet all applicable requirements of this specification shall be used.

3.4 Temperature requirements. All mechanical movements shall remain free and shall function satisfactorily without readjustment or change of lubricant when the instrument is in thermal equilibrium at any ambient air temperature between 120° F and minus 25° F is exposed to the full impact of solar radiation of 360 B.t.u./sq ft/hr for not less than 4 hours at an ambient air temperature of 125° F, and is exposed to minus 25° F temperature for not less than 3 days without benefit of solar radiation. Unless otherwise specified (see 6.2), the transit and accessories shall not be damaged by storage at any ambient temperature from 155° F for a period of not less than 4 hours to minus 65° F for a period of not less than 3 days.

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TABLE I. Dimensions and Physical Characteristics.

Class	Telescopic Magnification Diameters	Weight[1] Maximum	Circle Diameter[2]		Minimum Shift of Instruments on Foot Plate	Minimum Diameter of Exit Pupil	Minimum Field of View	
			Horizontal	Vertical				
		(lb)	(in)	(in)	(in)	(mm)	(deg)	(min)
A	16-21	11	4-1/2 to 5-1/8	4 to 5	5/8	1.50	1°	20'
B	19-24	14-1/2	5-1/4 to 6	4-1/2 to 5	5/8	1.40	1°	10'
C	22-28	17-1/4	6-1/8 to 7	4-3/4 to 5-1/4	3/4	1.20	1°	0'

1 Weight of complete transit equipped with sunshade but without plumb bob or auxiliary equipment (see 3.16).

2 The circle shall be measured at the edge of the graduations where readings are taken with the verniers while angles are measured.

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3.5 Performance. The instrument shall be easily and accurately adjustable and shall retain this ease and accuracy under the severe conditions normally encountered during service in all climates and regardless of the aging, warping, and internal strains of the materials. In order to insure ease of adjustment, screws and nuts for adjusting levels, reticle, trunnion bearing block, vertical circle vernier, and tangent screw collars shall be of the capstan type. The limbs and verniers shall resist damage by dust particles equally as well as those made with sterling silver overlays.

3.6 Overall accuracy. In any normal surveying operation in the field or as reproduced in the laboratory, the instrument shall consistently and repeatedly measure horizontal and vertical angles between lines of sight with an error not to exceed 30 seconds; it shall consistently and repeatedly measure vertical angles from the horizon as defined by the plate levels with an error not to exceed 45 seconds and shall produce a horizontal line by means of the telescope vial with an error not to exceed 15 seconds. The collimation error of the line of sight shall not be perceptible. Such accuracy shall be maintained from single readings in air temperatures varying from 15° to 120°F.

3.7 Telescope. The telescope shall be of the internal-focusing type. The telescope shall transit through the full vertical circle with prismatic eyepiece attached. The telescope and trunnion parts shall be so constructed that with the instrument adjusted for normal use and with all clamps set, the vertical hair will not move appreciably off a vertical line when the telescope tangent screw is moved back and forth. The vertical axis of the instrument shall be suitably marked on the top of the telescope.

3.7.1 Aberration. The system shall produce a sharp image throughout the entire focusing range and shall be sensibly free from spherical aberration, chromatic aberration, astigmatism, coma, and distortion. The field of view shall be substantially flat from the center to a zone outside the stadia wires equal to one-quarter of the stadia interval and shall show no marked curvature from there to the edge of the field. The system shall be sensibly free from internal reflections.

3.7.2 Eyepiece. Unless otherwise specified (see 6.2), the eyepiece shall be of the erecting type having a mounting of dull dark color to reduce reflection and shall have a suitable eyepiece shutter.

3.7.3 Minimum focus. The shortest focus shall be not greater than 6 feet measured along the optical axis from the vertical axis of the instrument.

3.7.4 Lens coating. All internal lens-air surfaces shall be coated with a low-reflecting coating conforming to Military Specification MIL-C-675.

3.7.5 Resolution. The limiting angle of resolution shall be not greater than 6 seconds.

3.7.6 Reticle. The reticle and supporting parts shall be such that distortion will not be introduced when collimation is being adjusted. The hairs shall be not less than 0.00013 inch nor more than 0.00020 inch in width; the hairs shall not exhibit noticeable unevenness and shall

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appear opaque. Hairs shall be either of platinum or glass fibre or shall be registered on optical glass. Unless otherwise specified (see 6.2), the hairs shall consist of a horizontal hair and two stadia hairs parallel to the horizontal hair and equidistant from it and a vertical hair perpendicular to the other three. When a glass reticle is furnished, the stadia hairs shall extend right and left of the vertical cross hair a distance of 5 minutes of true field, plus or minus 20 percent. Stadia hairs shall provide the horizontal distance from the vertical axis of the instrument to the rod, without allowance for constant, to plus or minus 0.3 percent for all distances greater than 100 feet. Unless otherwise specified {see 6.2), the stadia ratio shall be 1:100.

3.8 Standards. The standards shall be strong and rigid and ample bases for properly mounting the telescope and all auxiliary equipment specified in 3.16 and shall be securely mounted on, or integral with, the plate.

3.9 Horizontal circle.

3.9.1 Graduation. Graduations on the circle and verniers shall have a contrast at least equal to that of black-filled lines on sterling silver with matte finish. Surfaces shall be uniform and free from porosity. The design shall be such that when even minutes are observed from a vertical angle of 45 degrees, the coincident graduation lines on the plate and the vernier shall appear straight, continuous, clearcut, and uniform in width. Lines shall be cut to sufficient depth to permit a reasonable amount of regrinding and reconditioning of the surfaces. The horizontal circle shall be graduated to 30 minutes and numbered every 10 degrees. Numbers shall be inclined in the direction of increase. Unless otherwise specified (see 6.2), the inner *rows* of numbers shall increase clockwise from O degrees to 360 degrees, and the outer row shall increase counterclockwise from 0 degrees *to*

3.9.2 Verniers. There shall be two direct-type double opposite verniers offset approximately 30 degrees from the line of sight so that they may be easily read. Verniers shall be graduated so as to read directly to 1 minute and shall be suitably numbered at 10-minute intervals. The vertical distance between the graduated surfaces of the verniers and limb combined with the horizontal distances between the vernier and limb shall be such that parallactic error will not exceed 30 seconds when viewed at a vertical angle of 45 degrees and a horizontal angle of 15 degrees. The vernier cover glass shall be set in the cover plate and shall be sealed waterproof. Unless otherwise specified (see 6.2), each vernier shall be equipped with a nonflammable reflector. The centering of the limb and the setting of the verniers shall be such that in any position, the reading of one vernier minus that of the other shall not vary from 180 degrees by more than 15 seconds.

3.10 Vertical circle. Unless otherwise specified (see 6.2), a full vertical circle, truly concentric with its mounting, having its graduated surface in the same plane as that of the vernier and with a rigid guard designed to protect the graduation, shall be provided.

3.10.1 Graduations. Graduations on the circle and verniers shall have a contrast at least equal to black-filled lines on sterling silver with matte finish. The surface shall be free from porosity. The lines shall be cut to

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sufficient depth to permit a reasonable amount of regrinding and reconditioning of the surfaces. Unless otherwise specified (see 6.2), the vertical circle shall be graduated to 30 minutes numbered every 10 degrees; the numbering shall be quadrants with the zero designations parallel to the vertical axis of the instrument; the vernier shall be graduated so as to read directly to 1 minute and shall be suitably numbered at the lines indicating 10-minute intervals.

3.10.2 Vernier. Unless otherwise specified (see 6.2), one direct-type double vernier shall be furnished. Means for adjusting and holding the vernier in its proper position shall be provided.

3.10.3 Stadia arc. When specified (see 6.2), the vertical circle shall contain a stadia arc. The stadia arc shall consist of separate horizontal and vertical scales and appropriate indices so that stadia readings may be reduced to horizontal distances and differences of evaluation by simple multiplication. Means for adjusting and holding the indices in their proper position shall be provided.

3.10.4 Control level. When specified (see 6.2), a control level vial that is adjustable by tangential motion shall be furnished.

3.11 Compass. Unless otherwise specified (see 6.2), the transit shall be supplied with a magnetic compass. The compass box shall be rainproof and dustproof; the cover plate shall be readily removable to permit balancing of the needle. There shall be a suitable device for damping and raising the needle from the pivot. Convenient means shall be provided for setting off the magnetic declination to at least 25 degrees in either direction without removal of the cover plate. The interior of the compass box exclusive of the graduated compass circle shall have a nonglare finish. The compass circle shall be truly concentric with the pivot and shall be graduated by degrees and half-degrees, and numbered every 10 degrees (O degrees to 90 degrees) within each quadrant, with the zero graduations parallel to the line of sight of the telescope; the graduations shall have a contrast equal to that of black-filled lines on sterling silver with matte finish. The compass needle shall be symmetrically magnetized and shall be not less than 2-7/8 inches long. The needle shall be designed and constructed so as to assure long retention of magnetism. There shall be no apparent differences in the reading of the two ends of the needle in any position on the compass circle when viewed with the magnifying glass accessory. The needle shall be mounted with a setting or “jewel,” polished smooth inside and without cracks; the needle shall be freely supported on a pivot of hardened steel sharpened to a smooth needle point. The needle shall be provided with a balance weight which can be moved readily to any position on either end to assure proper balance in any latitude in either hemisphere. The center of gravity of the needle shall be near the point of support so that there is a slight tendency for the needle to quiver. The needle shall be balanced when demagnetized and without the balance weight.

3.12 Level vials. All level vials shall be graduated to 2-millimeter spacing. The graduation shall be clear, wear-resistant, uniform, symmetrical about the exposed portion, and shall extend so as to include the end of the bubble at all temperatures from minus 25° F to 120° F. All vials shall be ground to uniform curvature within 10 percent of rated sensitivity. Vials shall be mounted with provision for adjustment. Vials and mountings liable to damage by the telescope shall be guarded. Unless otherwise specified

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(see 6.2), the plate vials shall be ground to a sensitivity of between 60 and 90 seconds; telescope vials shall be *ground to* a sensitivity of between 20 and 40 seconds. When specified (see 6.2), the telescope level vial shall be of the reversion type.

3.13 Clamps and tangent screws. With the telescope set on a definite point, the tightening of any clamp screw shall not move either cross hair more than 20 seconds and no movements of plate level bubbles shall be evident. All tangent screws shall be 1/4 inch in diameter, 40 threads per inch (1/4-4ONS) conforming to FED-STD-H28 and shall provide a smooth and uniform movement of at least 3 degrees of arc. All clamp and tangent screws shall be conveniently placed and readily accessible. The head of the lower clamp screw and the head of the lower tangent screw shall be distinctively different in design. Provisions shall be made for wear adjustment or individual screw and assembly shall be replaceable.

3.14 Leveling head and base plate. Design and workmanship shall be such as to provide leveling with ease and facility and without cramping or shifting when the base plate is within 7 degrees of level. The leveling screw design shall be such that smooth universal action will be assured and the shoes will not in themselves become detached. Provisions shall be made for wear adjustment or individual leveling screw assemblies shall be replaceable. Threads of leveling screws shall be completely inclosed. Leveling screw threads shall be 3/8 inch in diameter, 32 threads per inch (3/8"-32NS) conforming to FED-STD-H28. The base-plate thread shall be 3-1/2 inches-8 in accordance with figure 1, and the plate shall be marked “3-1/2 x 8.” A device for attaching a plumb bob shall be provided. Unless otherwise specified (see 6.2), a four-screw leveling head shall be provided.

3.15 Standard accessories. Unless otherwise specified (see 6.2), the following accessories shall be furnished with each instrument and packed in the proper fixture in the carrying case:

Item	Quantity
Cover, waterproof	1
Magnifier, conforming to Federal Specification GG-M-95 type II, class A, 3.5 power, 30 mm lens diameter	1
Pins, adjusting	2
Screw driver	1
Wrenches as required for field stripping.	
Plumb bob, 16 ounce with cord and adjuster, conforming to Federal Specification GGG-P-501, type I, class B	1
Sunshade	1
Cap, objective-lens	1
Brush, lens-cleaning	1
Booklet, instruction on maintenance care and adjustment	1
Eyepiece, prismatic, with filter for solar observation	1
Reflector, sunshade type, for illuminating cross hairs at night	1
Reticle, spare	1
Level vials, mounted in tubular metal housing (one replacement for each level)	3

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3.15.1 Military lubricant. Transits purchased for Military use shall be lubricated with grease conforming to Military Specification MIL-G-10924, and a container filled with not less than 1/4 ounce of military lubricant shall be furnished with each instrument.

3.16 Auxiliary equipment. An auxiliary telescope or a solar attachment shall be furnished when and as specified (see 6.2).

3.17 Tripods. Tripod legs shall be of white ash, beech, yellow birch, hickory, hard maple, or pecan. Slope of grain in wood shall not exceed 1 in 15, and moisture content shall be not more than 12 percent. Wood shall be free of decay, bark or gum pockets, splits, holes, unsound or loose knots, wane, warp, sound knots exceeding 1/4 inch in diameter, knot clusters, and surface checks or shakes except those of barely perceptible opening not more than 4 inches in length. The shoes shall be of hard metal, and all other metal parts shall be of strong nonferrous metals such as brass or bronze. The tripod heads shall have a male thread 3-1/2 inches-8 in accordance with figure 1; it shall be marked "3-1/2 x 8," and shall be provided with a metal protecting cap. A sturdy pointed tripod shoe shall be firmly fastened to the lower end of each leg. When specified (see 6.2), suitable push-down spurs shall be furnished. The tripod legs shall be attached to the head by means of hinges which shall have good lateral rigidity, the connections being not less than 1-1/4 inches wide. Each hinge shall be strong, free from play, and shall be capable of being tightened easily. All parts shall include adequate provision against loosening and loss in service while permitting removal for repairs. The hinges shall be firm enough so that, when tightened, each tripod leg will sustain its own weight when held at a 45 degree angle. Overall length of the tripods with legs placed together and with the cap of the tripod head removed shall be not less than 58 inches. If the tripod has extension legs, the measurement is to be made with the legs fully extended. A binding strap rigidly attached to one leg shall be provided. Tripods shall weigh not less than 10 pounds or more than 13 pounds.

3.17.1 American type. The tripod head shall be of one piece having three sets of legs with machined faces not less than 1-1/2 inches in diameter. Tripod bolts shall be not less than 3/8 inch in diameter and shall be provided with a means of preventing the bolt from turning when the nut is being tightened. A washer shall be placed under the wing nut.

3.17.1.1 Extension legs. Each leg shall be equipped with two clamps of ample capacity, securely fastened to the leg so that the extension may be clamped and held firmly in position. Clamp screws shall be of the wing-nut type with plates to prevent the screws from cutting into the wood. Telescoped length shall be not more than 36 inches. When a fixed length tripod is specified, these requirements do not apply.

3.17.2 European type. Each tripod leg shall be attached to the head by means of metal lugs fixed to the leg; such lugs shall be not less than 3/8 inch in diameter and 1/2 inch in length. The lugs shall be snugly fitted into cylindrical bearings in the head and shall be provided with a means of tension adjustment.

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3.17.2.1 Extension legs. Extension legs shall conform to 3.17.1.1, except that one clamp on each leg shall be permitted and the telescoped length shall be not more than 41 inches.

3.18 Carrying case. Unless otherwise specified (see 6.2), the transits shall be well packed in a sturdy varnished case with mortised joints and comparable in durability to Honduras mahogany. The case shall be designed so that the transit will not be affected by ordinary handling, but may be readily packed and unpacked for use. The case shall be equipped with rubber bumpers on the bottom, hook, lock, and hinges of homogeneous noncorrosive metal. Unless otherwise specified (see 6.2), a pack harness and carrying strap of cotton webbing shall be furnished. Fittings shall be provided for holding all accessories and auxiliary equipment so that they will not be damaged or come in contact with one another or with the transit under ordinary handling. The design of the fittings shall be such that an operator wearing heavy gloves can quickly and conveniently remove and replace the objective cap, plumb bob, and sunshade. Cotton webbing for the carrying strap shall conform to Military Specification MIL-W-530, type III, *class 4*, 1-1/2 inches wide, olive drab No. 7. The ends of the strap shall be folded under 1-3/4 inches, stitched, and securely fastened to the case so that the transit can be carried in an upright and balanced position. Cotton webbing for the pack harness shall conform to Military Specification MIL-W-530, class 4, olive drab No. 7, type II for the 2-inch-wide webbing and type 111 for the l-inch-wide webbing. The harness straps shall be fitted with buckles conforming to MIL-B-543, type II, class 1, 1 inch (construction A or B); brass and clips conforming to MIL-C-496, type I, 1 inch; and snaps conforming to MIL-S-43770 class optional. The webbing shall be stitched and riveted where the loops are formed for snaps and buckles. The case shall be provided with metal ring holders designed, positioned, and securely fastened to the case. The ring holders shall be fitted with “D” rings conforming to MIL-R-3390, class optional, 3/4 inch by 5/8 inch, without roller, style 1 (symbol X). The case shall be constructed so that the transit can be removed by grasping the leveling head. Springs or pads of shock-absorbing material shall be provided for limiting the motion of the transit within the carrying case. When a wood case is furnished, metal case corners shall be securely fastened to the exterior comers. Metal parts of the carrying case and pack harness for which materials are not specified shall be made of corrosion-resistant metal or shall be treated to resist corrosion. Metal parts on the exterior of the carrying case shall have a dull finish or the parts shall be given one coat of enamel conforming to TT-E-522 color optional, after surface treatment to insure adhesion of the enamel.

3.19 Identification marking. The equipment shall be identified in accordance with MIL-STD-130. Other markings shall be as specified (see 6.2).

3.20 Treatment and painting. Wood surfaces shall be finished, treated, and painted in accordance with MIL-T-704, type A. All other surfaces shall be finished, treated, and painted in accordance with the manufacturer’s standard practice for this type of equipment.

3.21 Technical publications to accompany equipment. Such technical publications as are specified shall be furnished (see 6.2).

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3.22 Items to be furnished with prime equipment. No shipment of prime equipment shall be made unless accessories are included with the shipment or unless written approval for shipment of the prime equipment without such items has been received from the contracting officer.

3.23 Workmanship. The transits shall have a uniform, unmarred finish. Each component shall be accurately machined or molded so that when put together they fit smoothly and move freely. Internal and external parts shall be clean and free from residues of fabrication.

3.23.1 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and otherwise to the manufacturer's standard practice. Finished contact and bearing surfaces shall be true and exact.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may we his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction model. The preproduction model shall be examined for the defects listed in 4.4 and tested as follows. Failure to pass the examination or any of the tests shall be cause for rejection.

4.2.1 Magnification. The magnification of the telescope shall be tested for conformance to table I by placing a low-power auxiliary telescope behind the eye lens of the transit telescope and focusing the transit telescope for parallel light. Using care not to change the focus, place a graduated transparent scale in front of and as near to the objective lens of the transit as practicable. Replace the low-power auxiliary telescope with a measuring microscope and focus on the image of the scale. The magnifying power of the telescope in diameters is the ratio of a given length of the scale to the length of the scale image in the exit pupil. The magnification of the transit telescope may be tested by any other appropriate method yielding values accurate within 2 percent.

4.2.2 Diameter of exit pupil. With the telescope adjusted, the microscope mounted on the micrometer slide is racked in and out until it is focused on the smallest sharply defined bright circle visible through the microscope. The measured diameter of this circle is the diameter of the exit pupil and shall conform to table 1.

4.2.3 Field of view. Set the vernier at 0 degrees. Point the telescope at a suitable target not less than 300 feet from the instrument. With lower motion, bring the target to the intersection of the horizontal diameter and edge of field of view. By upper motion, bring the target to the intersection of the horizontal diameter and opposite edge of the field of view. To determine conformance to table I, read the horizontal angle which equals the field of view.

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4.2.4 Minimum focus. Set the telescope at minimum focus. Move a graduated scale along the line of sight until its markings are most sharply defined. The distance from this position of the scale to the vertical axis of the instrument is the minimum focus and shall be checked for conformance to 3.7.3.

4.2.5 Resolution. Observations for resolution shall conform to 3.7.5 and shall be made within the stadia lines, employing charts contained in either NBS Circular 428 or 533. An observer with normal vision shall be capable of clearly resolving Pattern 28 and all coarser patterns with the chart placed 100 feet from the instrument.

4.2.6 Temperature tests.

4.2.6.1 Low temperature tests. The transit shall be subjected to a temperature of minus 25° F. for at least 3 days without benefit of solar radiation. All motions shall be tested to determine conformance to 3.4. The temperature shall then be lowered to minus 65° F. and maintained for 3 days. The instrument shall then be examined to determine conformance to 3.4.

4.2.6.2 High temperature tests. The transit shall be subjected to a temperature of 120° F. long enough to permit the largest internal mass in the instrument to attain temperature stability. While at this temperature, the instrument shall be exposed to a source of directed radiant energy simulating solar radiation of 360 B.t.u./sq. ft./hr. for a period of 4 hours. All motions shall be tested to determine conformance to 3.4. With the source of solar radiation removed, the temperature shall be raised to 155° F. and maintained for 4 hours. The instrument shall then be examined to determine conformance to 3.4. A suitable source of radiant energy to be used in the above test is five 300-watt R-40 reflector spot incandescent lamps (or equal) mounted on 15-cm. (6-inch) centers in the form of a cross with one lamp at the center common to both limbs. Such a source, with the plane of the cross horizontal and mounted directly over the transit at a height of 96.5 cm. (33-inch) from the tip of the center lamp to the top of the cover plate of the transit, will give the required radiant flux to a close approximation. If a radiation meter is available (see 6.3), the height of the source shall be adjusted so that the value of the flux is 1.5 gram calories per minute per square centimeter when the meter is placed in the positions to be occupied by the periphery of the cover plate of the instrument to be tested.

4.2.7 Spacing of reticle hairs. The spacing of reticle hairs shall be tested by an appropriate method to determine conformance to 3.7.6. One method of testing is to use a reticle with hairs correctly spaced and superimposed upon the cross-hairs of the instrument being tested.

4.2.8 Overall accuracy. The transit shall be given the tests specified herein to determine conformance to 3.6.

4.2.8.1 Horizontal circle. Either the field test or the laboratory test specified in 4.2.8.1.1 and 4.2.8.1.2, respectively, may be used.

4.2.8.1.1 Field test. The field test shall consist of laying off a series of angles so chosen that the overall accuracy can be determined. Set a line target at point A and horizontal scale targets at points B and C so that the angle at the instrument between A and B is close to 100 degrees and the angle at the instrument between A and C is close to 140 degrees. Each

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target should be measured 171.8 feet from the instrument. Carefully set the A-vernier at O degrees 05 minutes using the 5-minute graduation for coincidence and the graduation on each side for equal difference observations. Point the instrument at A; turn off the selected angles (close to 100 degrees and close to 140 degrees) on the plate, setting the A-vernier by coincidence as above, and read the pointings on the B and C targets, respectively; repeat using initial settings of 30°10′, 60°15',90°20', 120°25', and 150030′. Next, obtain a similar set of pointings with the telescope reversed but making all settings on the B-vernier, beginning with 180005′, 210010′, 240015′, 270020′, 300025′, and 330030′, respectively. Twelve pointings have thus been obtained from each of the two selected angles. At the distance specified (171.8 feet), 30 seconds at the instrument will introduce a difference of three-tenths inch at the target. Recheck all doubtful values, observing carefully to determine whether the grouping of the results may indicate the instrument was not sufficiently adjusted. Compute the average pointing; the extreme pointings should not be farther removed from the average pointing than three-tenths of an inch for conformance to 3.6.

4.2.8.1.2 Laboratory test. The laboratory test to determine conformance to 3.6 will be conducted along the same basic lines as the field test except that other surveying instruments or suitable collimators will be used at targets A, B, and C so as to reduce the test area required to the limits of a small room. The cross hairs of a l-minute transit whose telescope is focused at infinity and pointed to the instrument under test may be used for target A. Precise transits, preferably reading to 5 seconds or better, may be similarly used for targets B and C. The difference between the first pointing and successive pointings may be determined directly on targets B and C by noting on the vernier or micrometer the number of seconds required to point the cross hairs of the target on the cross hairs of the instrument under test.

4.2.8.2 Vertical circle. The vertical circle shall be tested with a circle testing machine or by any method yielding results within the accuracies specified in 3.6.

4.2.8.3 Collimation. The effect of the focusing arrangement on the line of collimation shall be checked as follows. Point on a fine object at infinite focus and read the horizontal circle using both verniers; reverse the instrument, turn the upper motion, point and read again; find the quantity which represents the difference between the deflection angle and 180 degrees; repeat this operation pointing on a fine mark at close focus; the difference of these two values represents the combination of the change in the line of sight due to focusing and the offset of the line of sight over the vertical axis. The results of this test shall be examined to determine conformance to 3.6.

4.2.9 Magnetic materials (applicable only) to transits provided with compasses). Set up the transit, plumb it over a fixed point, and read the bearings of four targets approximately 90 degrees apart by means of the compass dial and the horizontal circle. Form the differences between each pair of readings as read on the compass circle and as read on the horizontal circle of the instrument. If corresponding differences formed by readings on the two circles do not differ more than one-half degree, the upper rotatable part of the transit is satisfactorily free from magnetic material and conforms to 3.3. Reset the transit over the same point as the preceding test but with tripod and lower stationary part of the instrument rotated through approximately 90 degrees. Take magnetic bearings on the four targets of the first

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part of test. If the magnetic bearings of parts 1 and 2 for any given target do not differ by more than one-half degree, the lower part of the transit is satisfactory free from magnetic material and conforms to 3.3.

4.2.10 Sensitivity of level vials. Test for conformance to 3.12 as follows: (1) Measure and reduce to angular values the distance on a vertical rod subtended by the movement of the level bubble of one division on the tube; or (2) mount the vial on the hinged beam, one end of which is moved vertically by means of a micrometer screw, each revolution of which results in a known angular movement of the beam.

4.2.11 Level vial graduations wear resistance. Each level vial to be tested for wear resistance of graduations shall be placed in a concentrated solution of either sulphuric acid or toluene at 100° F. for a period of ten days. At the end of the test period, the vial shall be examined for conformance to 3.12.

4.3 Sampling. Sampling for examination and tests shall be in accordance with Military Standard MIL-STD-105.

4.4 Examination. Samples selected in accordance with 4.3 shall be examined as follows. Lot acceptance or rejection shall be based on an AQL of 2.5 percent defective for major defects and an AQL of 6.5 percent defective for minor defects.

Major

1. Materials not as specified.
2. Instrument not easily adjusted (see 3.5).
3. Telescope does not transit thru full vertical circle (see 3.7).
4. Vertical axis of the instrument not suitably marked on the top of the telescope (see 3.7).
5. The telescope does not conform to the aberration requirements of 3.7.1.
6. Lens-air surfaces not coated as specified (see 3.7.4).
7. Reticle not permanent, accurately and evenly placed, opaque or of uniform width (see 3.7.6).
8. Standards not as specified (see 3.8).
9. Horizontal and vertical circle graduations not as specified (see 3.9.1 and 3.10.1).
10. Horizontal and vertical circle verniers not as specified (see 3.9.2 and 3.10.2).
11. Compass not as specified (see 3.11).
12. Level vials not provided with a means of adjustment or graduations not clear and uniform (see 3.12).
13. Clamps and tangent screws do not operate as specified (see 3.13).
14. Leveling screws do not operate as specified (see 3.14).
15. Base plate thread not as specified (see 3.14).
16. Accessories not provided or not as specified (see 3.15).
17. Auxiliary equipment not as specified (see 3.16).
18. Tripod head thread not as specified (see 3.17).
19. Workmanship not as specified (see 3.23).

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Minor

1. No eyepiece shutter (see 3.7.2).
2. Level vials not protected against damage (see 3.12).
3. Tripod protecting cap not provided (see 3.17).
4. Tripod not as specified (see 3.17).
5. Carrying case not as specified (see 3.18).
6. Equipment not marked as specified (see 3.19).
7. Wood surfaces not finished, treated, or painted as specified (see 3.20).
8. Equipment exceeds specified weight (see 3.1 and 3.17).

4.5 Acceptance tests. Samples selected in accordance with 4.3 shall be tested as specified in 4.2.7, 4.2.8, 4.2.9, and 4.2.10. Lot acceptance or rejection shall be based on an AQL of 2.5 percent defective. Samples failing to pass any of the tests shall be considered defective. The instrument shall be adjusted by any of the established methods prior to testing. The plate shall be considered level for all tests when the plate bubble is centered between graduations within 1/4 of a space.

4.6 Examination and tests of preparation for delivery. The packaging, packing, and marking of the transits and tripods shall be subjected to examination and tests to determine compliance with the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Commercial as specified (see 6.2).

5.1.1 Level A. The transits and tripod shall be packaged in accordance with Level A requirements of military specification MIL-S-3534.

5.1.2 Level C. The level C preservation for the transits and the tripod shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial preservation of the transits and the tripod shall be in accordance with the requirements of ASTM D-3951-82.

5.2 Packing. Packing shall be Level A, B, C or commercial, as specified (see 6.2).

5.2.1 Level A. The transits and tripods shall be packed in accordance with Level A requirements of Military Specification MIL-S-3534.

5.2.2 Level B. The transit and tripods shall be packed in accordance with Level B requirements of specification MIL-S-3534.

5.2.3 Level C. The level C packing for the transit and the tripod shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. The preserved transits and tripod shall be packed in accordance with the requirements of ASTM D-3951-82.

5.2.5 Palletized Loads. When specified (see 6.2), Palletized Loads, commensurate with the level of pacting specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be

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uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D-3951-82.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3 and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Intended use. The transits are intended for use in surveying operations. Class A covers an instrument of minimum weight, small dimensions, and compactness for use where those factors are of essential importance consistent with the required accuracy. Class B covers an instrument of medium weight and dimensions for use where transportation is not a critical problem or where circles larger than class A are desirable. Class C covers an instrument of maximum weight and dimensions for use where transportation facilities are readily available or where a higher order of refinement in determination of angular values is required.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Class of transit required (see 1.2).
(b) Type of tripod required (see 3.1 and 3.17).
(c) whether preproduction model is required (see 3.2).
(d) Whether storage temperature test is required (see 3.4).
(e) Whether telescope shall transit through the full vertical circle with prismatic eyepiece attached (see 3.7).
(f) Whether the eyepiece shall be of a type other than that specified (see 3.7.2).
(g) Whether reticle hairs shall be other than as specified (see 3.7.6).
(h) Stadia ratio required if other than as specified (see 3.7.6).
(i) Whether the graduation shall be other than as specified (see 3.9.1).

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(j) Whether each vernier shall not be equipped with a nonflammable reflector (see 3.9.2).
(k) Whether a vertical circle other than that specified shall be furnished (see 3.10).
(1) Whether the vertical circle shall be graduated other than as specified (see 3.10.1)
(m) Whether the type of vernier shall be other than that specified (see 3.10.2).
(n) Whether stadia arc is required (see 3.10.3).
(o) Whether a control level vial is required (see 3.10.4).
(p) Whether a compass is not required (see 3.11).
(q) whether the plate level vials shall be of a sensitivity other than as specified (see 3.12).
(r) Whether the telescope level vial shall be of the revision type (see 3.12).
(s) whether type of leveling head shall be other than that specified (see 3.14).
(t) Whether any of the accessories listed is not required (see 3.15).
(u) Whether auxiliary telescope and solar attachment shall be furnished and requirements for same (see 3.16).
(v) Whether push-down spurs shall be furnished (see 3.17).
(w) Whether the instrument shall be packed in a case other than as specified (see 3.18).
(x) Whether a pack harness and carrying strap is required (see 3.18).
(Y) whether marking other than that specified is required (see 3.19).
(z) Technical publications required (see 3.21).
(aa) Administrative provisions for inspection records (see 4.1).
(bb) Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial (see 5.1, 5.2 and 5.3).
(cc) When palletized loads are required (see 5.4).

6.3 A radiation meter such as the Type DW-60 (Model 8DW60Y2) of General Electric Company may be used to determine the radiation value specified in 4.2.6.2.

6.4 Figures 2, 3, 4, 5, and 6 show a type of transit, types of tripods, type of pack harness, and type of webbing which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other designs which conform to the specification.

6.5 Transportation description. Transportation description applicable to this item is:

Transits - Optical Instruments, NOIBN.
Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 10,000 pounds.

Tripods - Surveyor's
Carload minimum weight is 18,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 18,000 pounds.

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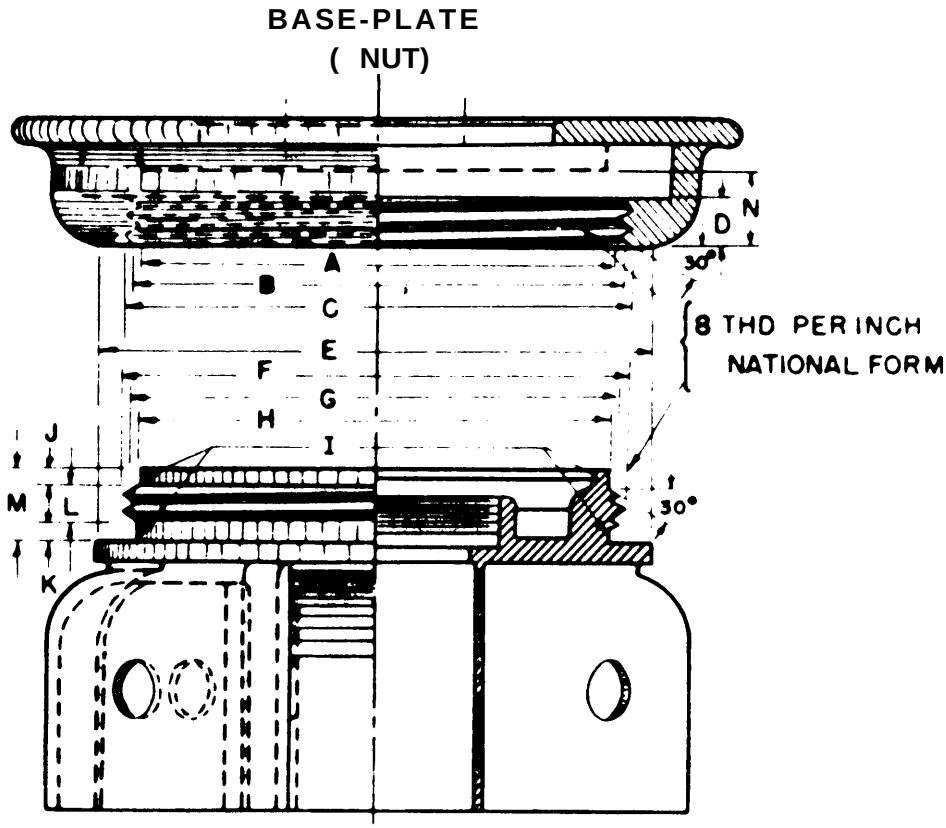
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BASE-PLATE NUT		MAX.	MIN.	TRIPOD-HEAD SCREW		MAX.	MIN.
MINOR DIAMETER	A	3.3792	3.3647	DIAMETER OF FLANGE	E		4 0000
PITCH DIAMETER	B	3.4232	3.4188	MAJOR DIAMETER	F	3 4966	3 4804
MAJOR DIAMETER	C	*	3.5000	PITCH DIAMETER	G	3 4154	3.4110
LENGTH OF TH'DED PART	D		15/32	MINOR DIAMETER	H	3 3432	3 3298**
FLANGE TO SHIFTING CENTER	N		17/32	DIAM. OF LEAD & GAIN	I	3 3298	
				LEAD	J	1/8	3/32
				GAIN	K	1/8	3/32
				LENGTH OF THREAD	L	11/32	1/4
				LENGTH OF THREADED PART	M	1/2	15/32

* CORRESPONDS TO FLAT EQUAL TO P ÷ 24 ** CORRESPONDS TO FLAT EQUAL TO P ÷ 8

FIGURE 1. -Thread for base-plate nut and tripod-head screw.

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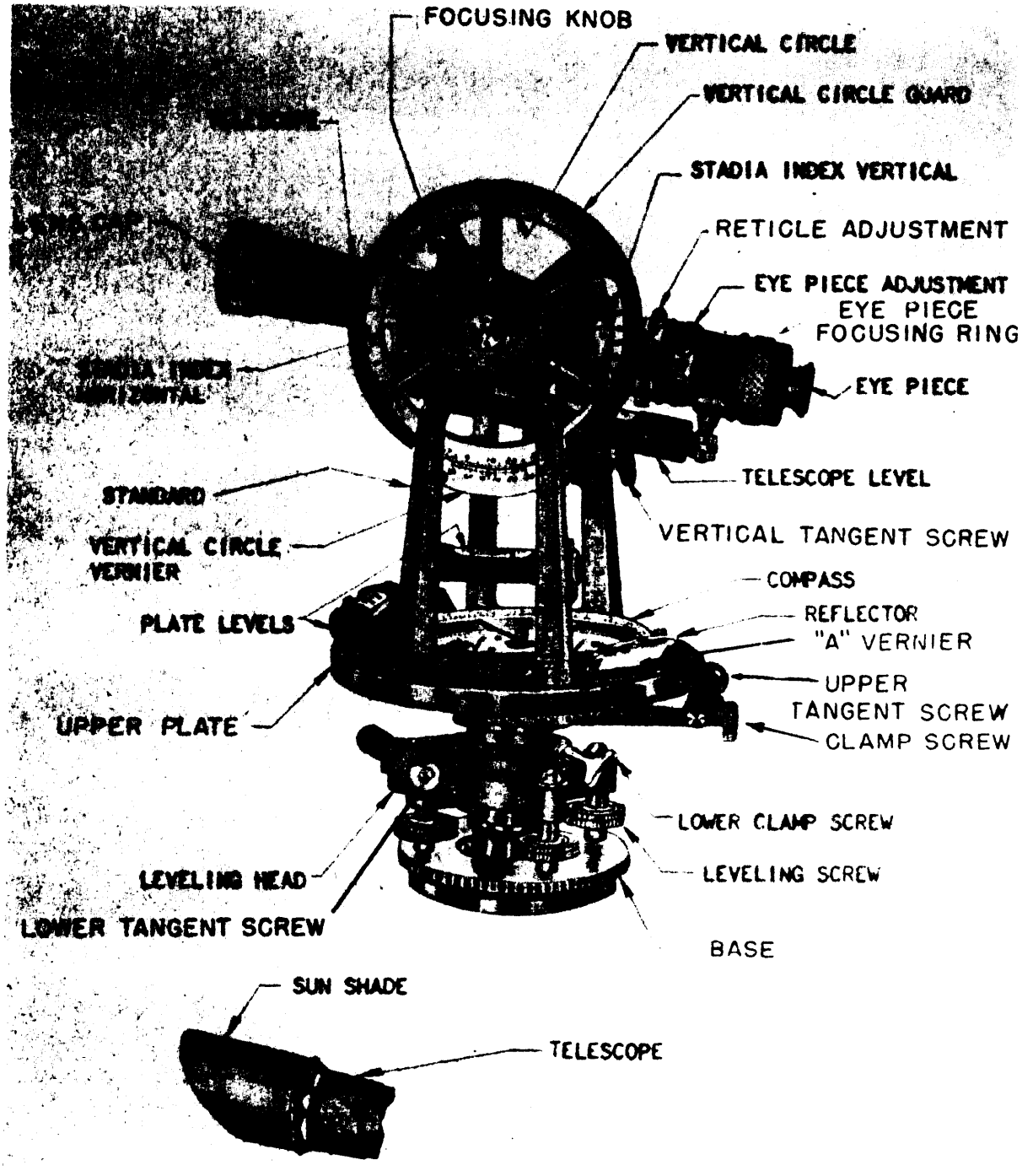


FIGURE 2.

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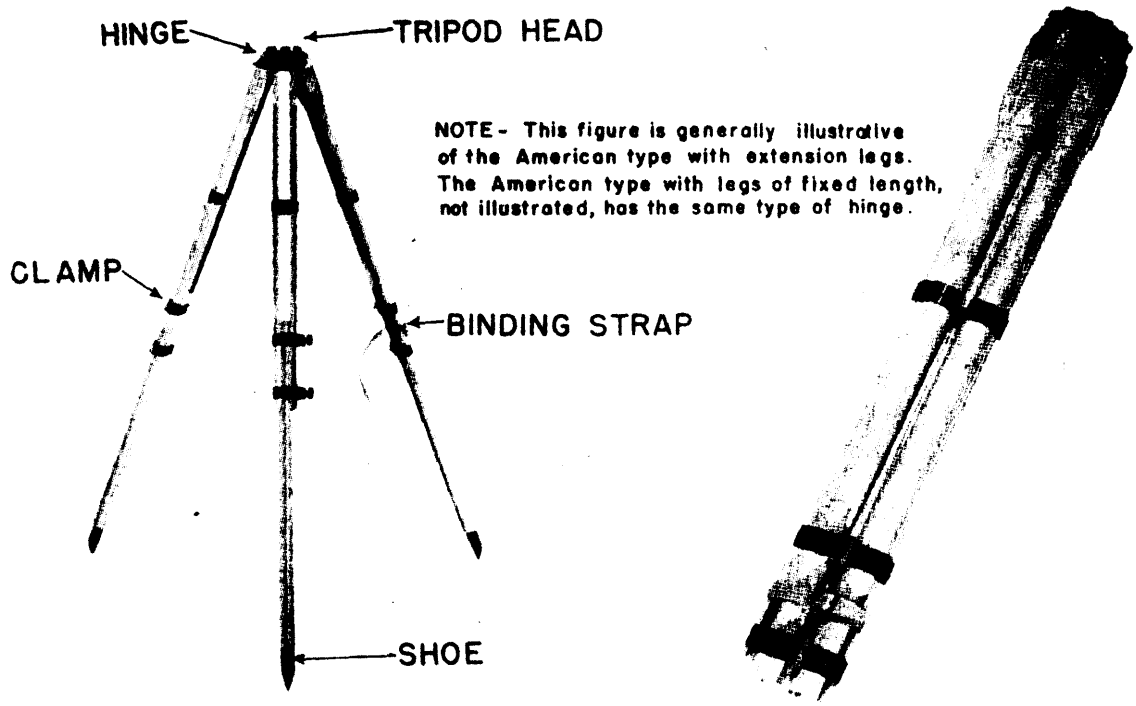


FIGURE 3. - American type tripod.

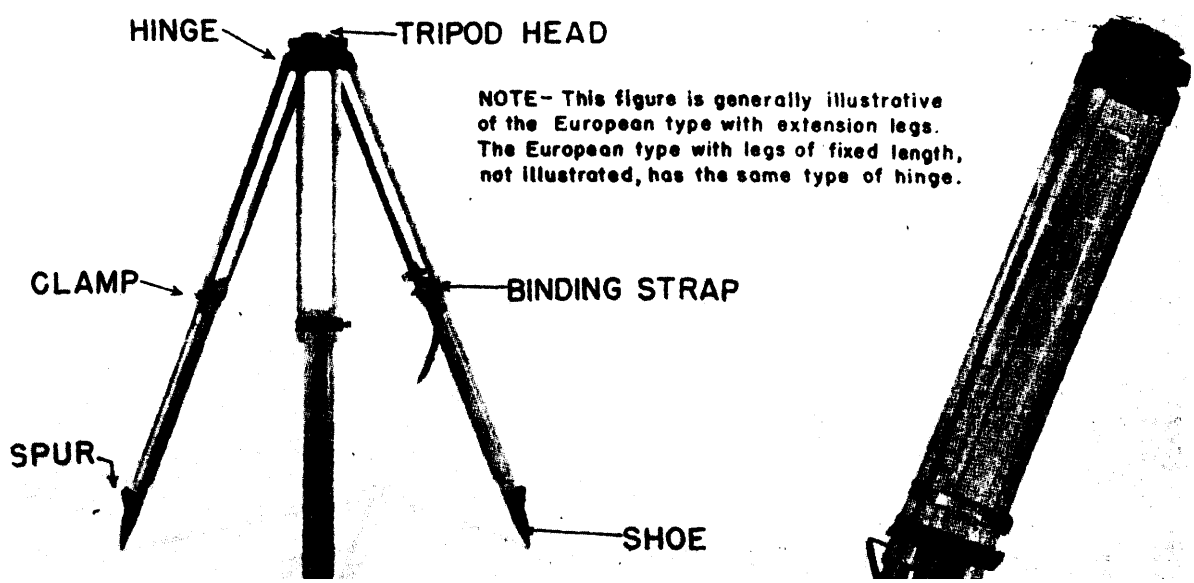


FIGURE 4.—European type tripod.

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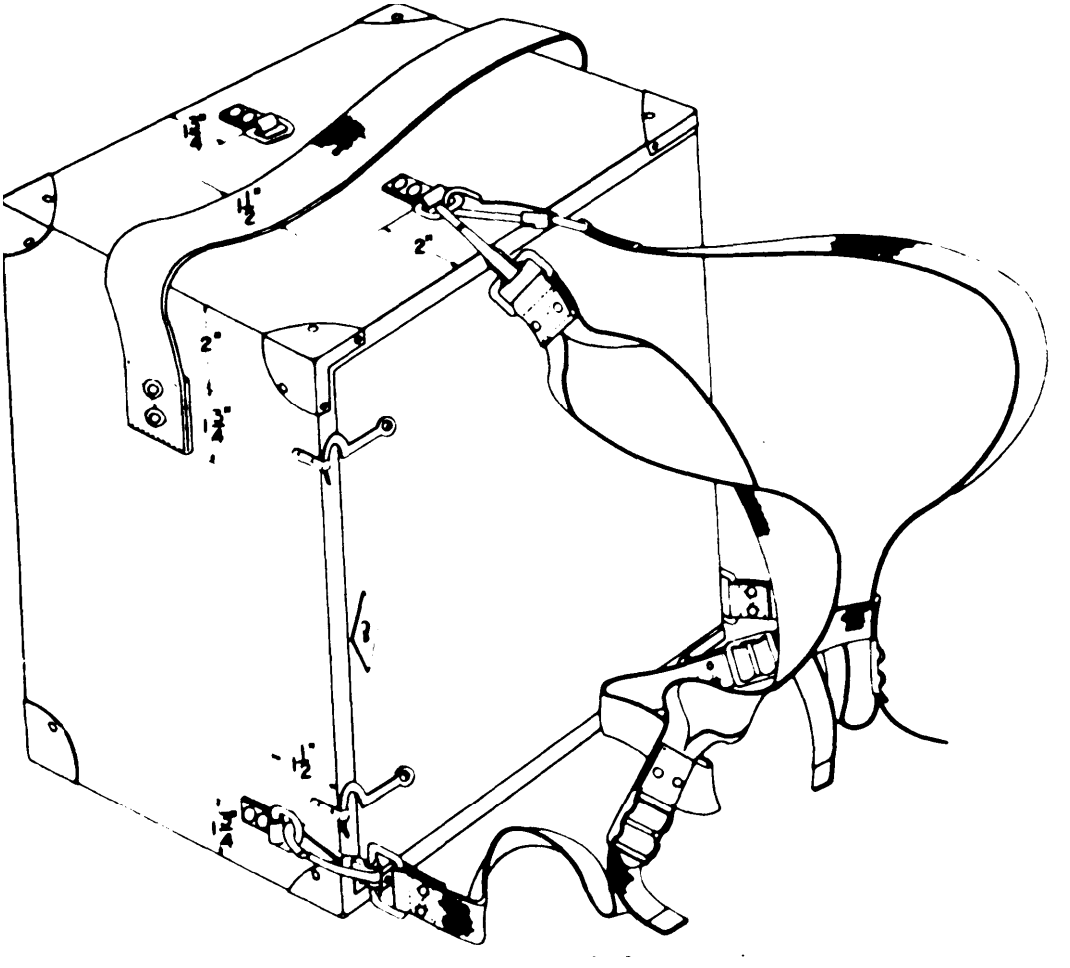


FIGURE 5. - Pack-harness attached to carrying case.

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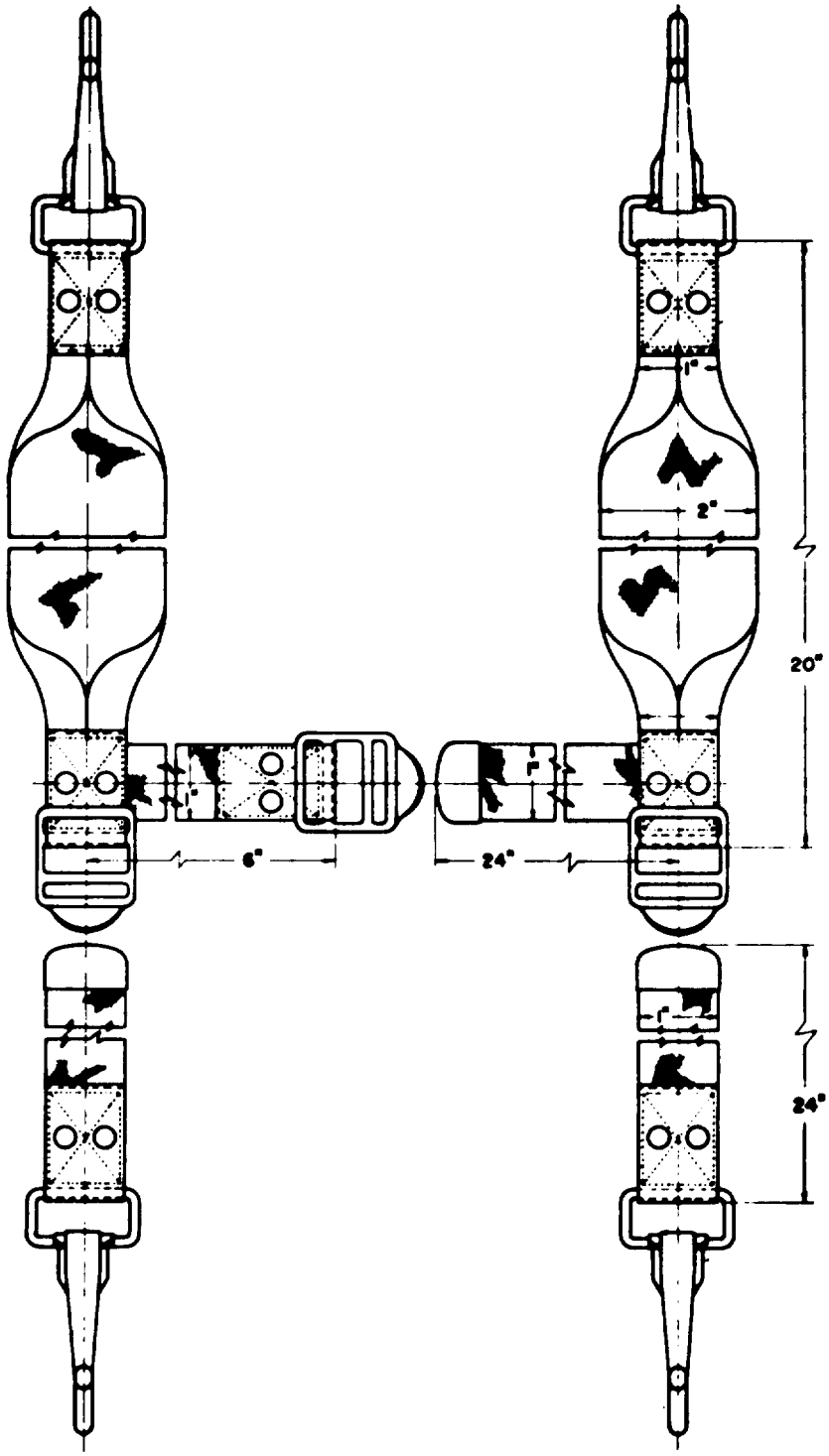


FIGURE 6.- Pack-harness.

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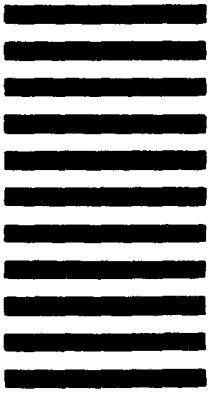
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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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